

CAMPUS® Datasheet

Grilon BG-30 S natural - PA6-GF30

EMS-GRIVORY | a unit of EMS-CHEMIE AG



Product Texts

Product-nomenclature: ISO 16396-PA6,GF30,M1HR,S14-100N

Markets

Automotive

Air intake sytems, Hydraulic systems, Automotive electr. and electronics, lighting, Interior

Electricals & Electronics

Electrical appliances, Electrical equipment

Industry & Consumer goods

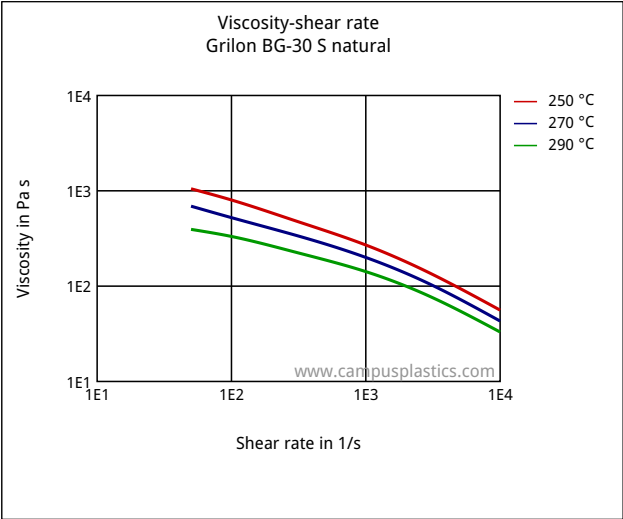
Housewares, Mechanical Engineering, Power transmission, Sports & Leisure, Tools & Accessories

Rheological properties	dry / cond	Unit	Test Standard
Molding shrinkage, parallel	0.1 / *	%	ISO 294-4, 2577
Molding shrinkage, normal	0.8 / *	%	ISO 294-4, 2577
Mechanical properties	dry / cond	Unit	Test Standard
Tensile modulus	9500 / 6000	MPa	ISO 527-1/-2
Stress at break	185 / 125	MPa	ISO 527-1/-2
Strain at break	5 / 10	%	ISO 527-1/-2
Charpy impact strength, +23°C	75 / 90	kJ/m ²	ISO 179/1eU
Charpy impact strength, -30°C	65 / 70	kJ/m ²	ISO 179/1eU
Charpy notched impact strength, +23°C	11 / 20	kJ/m ²	ISO 179/1eA
Charpy notched impact strength, -30°C	8 / 9	kJ/m ²	ISO 179/1eA
Thermal properties	dry / cond	Unit	Test Standard
Melting temperature, 10°C/min	222 / *	°C	ISO 11357-1/-3
Temp. of deflection under load, 1.80 MPa	205 / *	°C	ISO 75-1/-2
Temp. of deflection under load, 8.00 MPa	135 / *	°C	ISO 75-1/-2
Coeff. of linear therm. expansion, parallel	20 / *	E-6/K	ISO 11359-1/-2
Coeff. of linear therm. expansion, normal	110 / *	E-6/K	ISO 11359-1/-2
Burning behavior at thickness h	HB / *	class	IEC 60695-11-10
Thickness tested (h)	0.8 / *	mm	IEC 60695-11-10
Electrical properties	dry / cond	Unit	Test Standard
Volume resistivity	1E12 / 1E11	Ohm*m	IEC 62631-3-1
Surface resistivity	* / 1E12	Ohm	IEC 62631-3-2
Electric strength	40 / 37	kV/mm	IEC 60243-1
Comparative tracking index	- / 500	-	IEC 60112
Other properties	dry / cond	Unit	Test Standard
Water absorption	7 / *	%	Sim. to ISO 62
Humidity absorption	2 / *	%	Sim. to ISO 62
Density	1350 / -	kg/m ³	ISO 1183
Rheological calculation properties	Value	Unit	Test Standard
Density of melt	1200	kg/m ³	-
Thermal conductivity of melt	0.26	W/(m K)	-
Spec. heat capacity melt	2200	J/(kg K)	-
Eff. thermal diffusivity	9.85E-8	m ² /s	-
Ejection temperature	200	°C	-

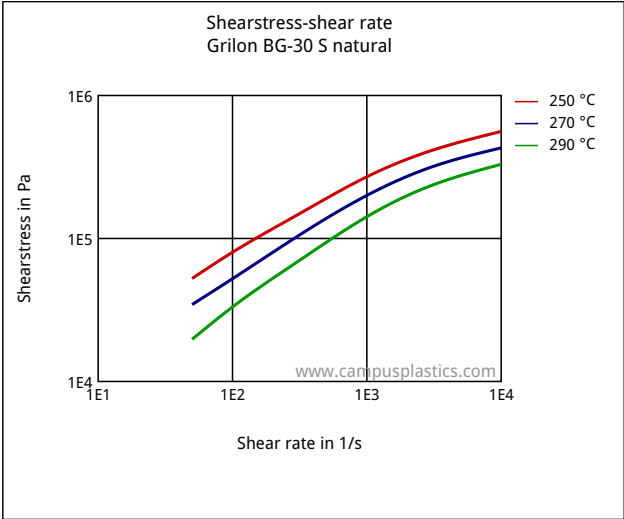
Test specimen production	Value	Unit	Test Standard
Injection molding, melt temperature	280	°C	ISO 294
Injection molding, mold temperature	80	°C	ISO 294
Injection molding, injection velocity	300	mm/s	ISO 294
Injection molding, pressure at hold	75	MPa	ISO 294

Diagrams

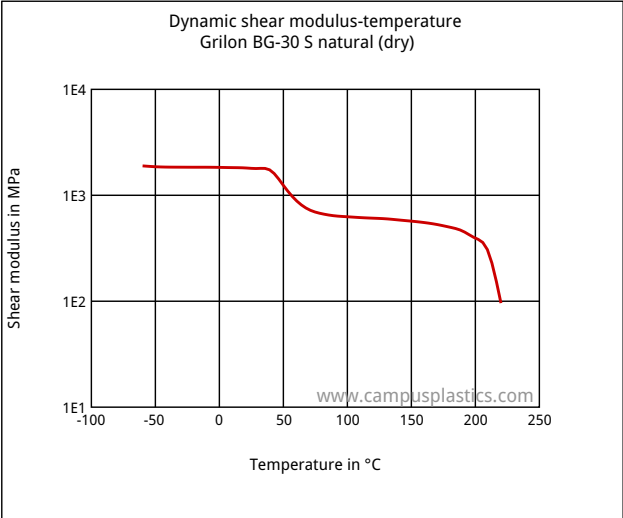
Viscosity-shear rate



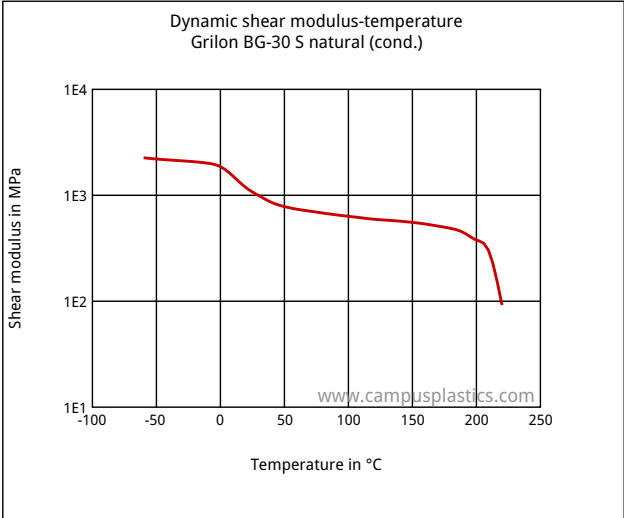
Shearstress-shear rate



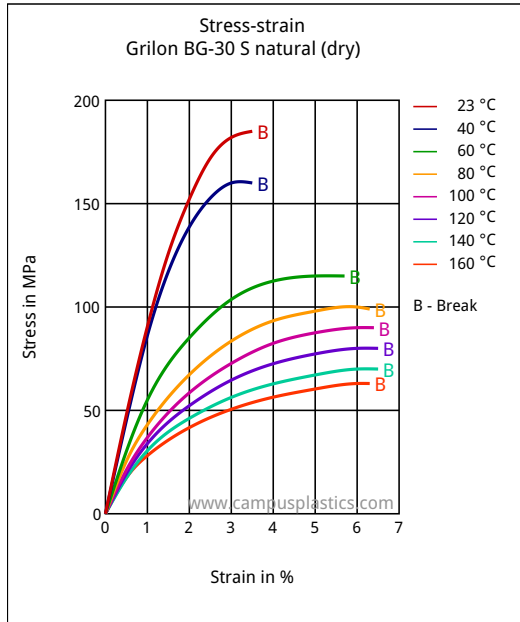
Dynamic shear modulus-temperature



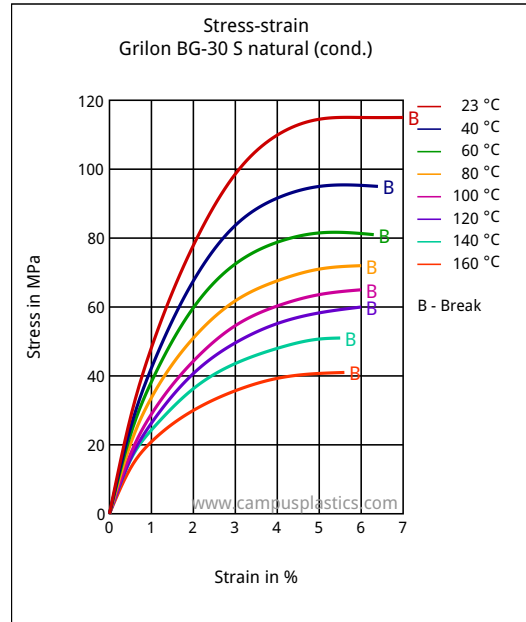
Dynamic shear modulus-temperature



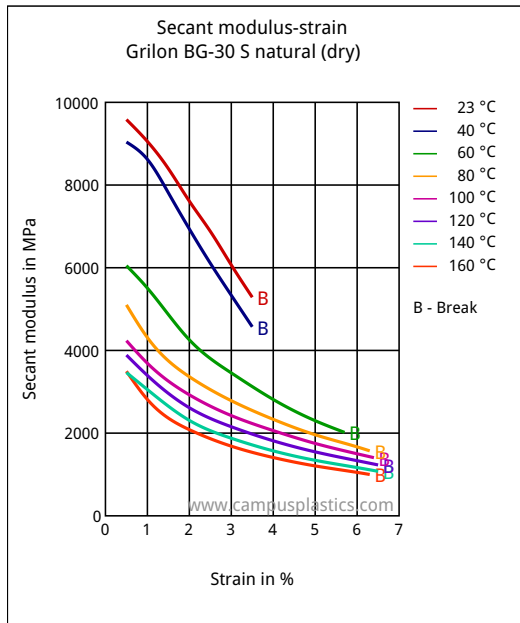
Stress-strain



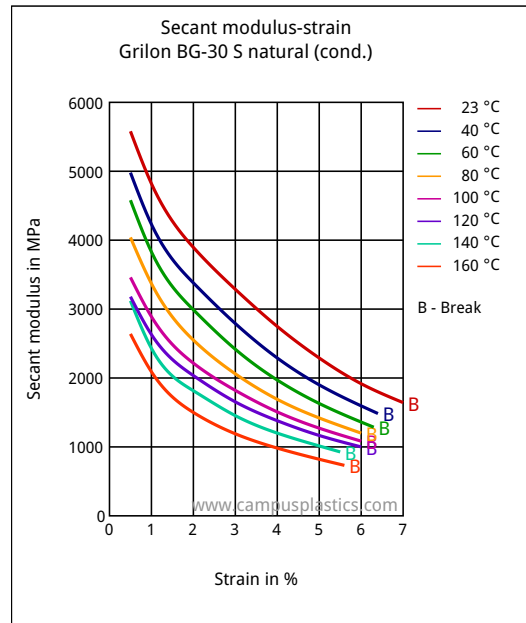
Stress-strain



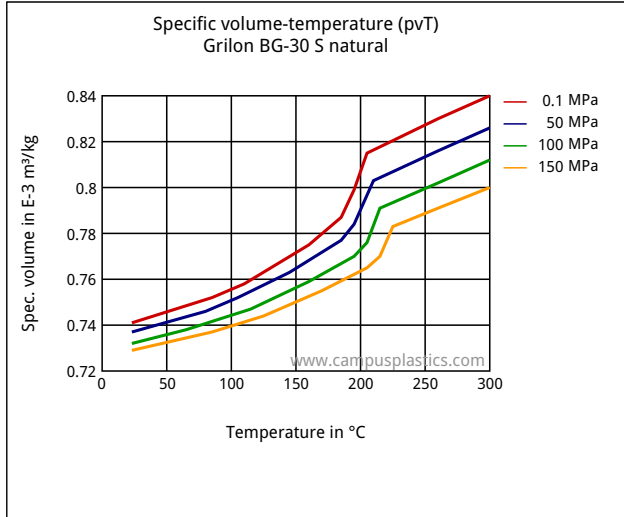
Secant modulus-strain



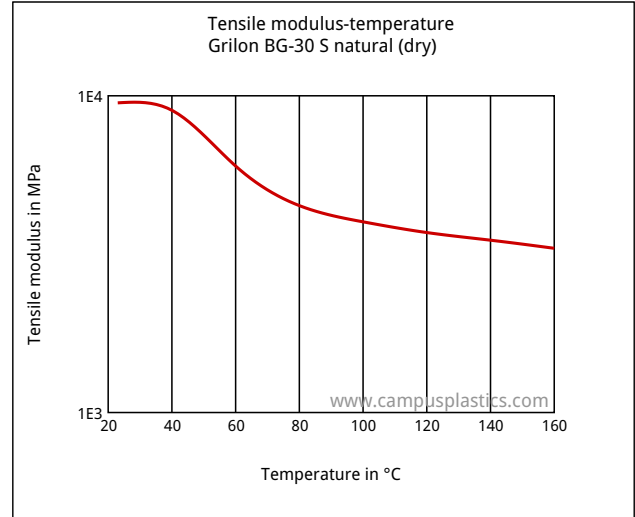
Secant modulus-strain



Specific volume-temperature (pvT)



Tensile modulus-temperature



Characteristics

Processing

Injection Molding

Delivery form

Granules

Regional Availability

North America, Europe, Asia Pacific, South and Central America, Near East/Africa

Chemical Media Resistance

Acids

- ☺ Acetic Acid (5% by mass) (23°C)
- ☺ Citric Acid solution (10% by mass) (23°C)
- ☺ Lactic Acid (10% by mass) (23°C)
- ☹ Hydrochloric Acid (36% by mass) (23°C)
- ☹ Nitric Acid (40% by mass) (23°C)
- ☹ Sulfuric Acid (38% by mass) (23°C)
- ☹ Sulfuric Acid (5% by mass) (23°C)
- ☹ Chromic Acid solution (40% by mass) (23°C)

Bases

- ☺ Sodium Hydroxide solution (35% by mass) (23°C)
- ☺ Sodium Hydroxide solution (1% by mass) (23°C)
- ☺ Ammonium Hydroxide solution (10% by mass) (23°C)

Alcohols

- ☺ Isopropyl alcohol (23°C)
- ☺ Methanol (23°C)
- ☺ Ethanol (23°C)

Hydrocarbons

- ☺ n-Hexane (23°C)
- ☺ Toluene (23°C)
- ☺ iso-Octane (23°C)

Ketones

- ☹️ Acetone (23°C)

Ethers

- ☹️ Diethyl ether (23°C)

Mineral oils

- ☹️ SAE 10W40 multigrade motor oil (23°C)
- ☹️ SAE 10W40 multigrade motor oil (130°C)
- ☹️ SAE 80/90 hypoid-gear oil (130°C)
- ☹️ Insulating Oil (23°C)

Standard Fuels

- ☹️ ISO 1817 Liquid 1 (60°C)
- ☹️ ISO 1817 Liquid 2 (60°C)
- ☹️ ISO 1817 Liquid 3 (60°C)
- ☹️ ISO 1817 Liquid 4 (60°C)
- ☹️ Standard fuel without alcohol (pref. ISO 1817 Liquid C) (23°C)
- ☹️ Standard fuel with alcohol (pref. ISO 1817 Liquid 4) (23°C)
- ☹️ Diesel fuel (pref. ISO 1817 Liquid F) (23°C)
- ☹️ Diesel fuel (pref. ISO 1817 Liquid F) (90°C)
- ☹️ Diesel fuel (pref. ISO 1817 Liquid F) (>90°C)

Salt solutions

- ☹️ Sodium Chloride solution (10% by mass) (23°C)
- 🚫 Sodium Hypochlorite solution (10% by mass) (23°C)
- ☹️ Sodium Carbonate solution (20% by mass) (23°C)
- ☹️ Sodium Carbonate solution (2% by mass) (23°C)
- ☹️ Zinc Chloride solution (50% by mass) (23°C)

Other

- ☹️ Ethyl Acetate (23°C)
- 🚫 Hydrogen peroxide (23°C)
- ☹️ DOT No. 4 Brake fluid (130°C)
- ☹️ Ethylene Glycol (50% by mass) in water (108°C)
- ☹️ 1% nonylphenoxy-polyethyleneoxy ethanol in water (23°C)
- ☹️ 50% Oleic acid + 50% Olive Oil (23°C)
- ☹️ Water (23°C)
- ☹️ Deionized water (90°C)
- 🚫 Phenol solution (5% by mass) (23°C)

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